

PHYSICAL ANTHROPOLOGY



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Book: Physical Anthropology (Schoenberg)

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Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

Preface

The high cost of textbooks prevents many students from succeeding. Textbook prices have gone up 3 times the rate of inflation in the past 30 years. New textbook editions are a scam. The change in Jurmain's *Introduction to Biological Anthropology* from 2009 to 2011 was to search and replace the word “hominid” for the word “hominin”, and raise the price 20%. Many students will not buy their textbooks (Kingkade 2013), and struggle through class, reading at the library. Other students will order cheap copies online but get the wrong edition, or discounted shipping where it arrives halfway through the semester. Textbooks have become a barrier to student success.

Assigning free online textbooks is one solution. Much of the high cost of textbooks comes from color printing, but black-and-white and text-heavy books are poor options for our predominantly visual learners. Online textbooks allow for unlimited, large, color graphics. For the price of a new textbook, a student can buy a text reader (kindle, ipad, tablet, surface, etc.) or a used laptop. There may still be problems with students' access to high speed internet. The monthly fee for a cable modem service is expensive, but free WiFi is becoming more and more widely available. The new downtown San Diego library is a good example.

Unfortunately, in 2011 when I started this, there were no free online textbook available for Introduction to Physical Anthropology. Probably the closest were Dennis O'Neil's Biological Anthropology Tutorials from Palomar College, Wikipedia's Biological Anthropology and Introduction to Paleoanthropology textbooks, and I borrowed from these heavily. I have found most of these sources written in the style of an encyclopedia, and I've tried to compensate for that, to make the textbook more approachable, by including my own voice whenever possible.

Some students may lack the necessary computer skills to use an online textbook. Hopefully the recent increases in funding for basic skills will help. Gone are the good old days whence we scribed our homework on tablets of wet clay. Students must learn computer skills to survive academically and professionally, and the printed textbook is becoming a relic of a bygone age. The more students practice current electronic research techniques, the better prepared they will be for the rest of their academic and professional life.

I previously encouraged students to use their textbook as the main source for my take-home tests, but I found that many students start with the internet as their first source, and then fall-back on the textbook if the info doesn't show up immediately in a search engine. I've come to realize that this is not always bad, as they often discover more current information than the textbook. For rapidly changing issues, such as how many genes are in the human genome, how closely we are related to Neandertals, *Homo naledi*, or the Cerruti Mastadon, the information available on sites such as Wikipedia is often more accurate than the typical textbook written three to ten years ago.

For each major section I have tried to follow a typical textbook chapter format:

- *Introduction: an introduction and short summary of the section.*
- *Focus questions: intended to help students consolidate the disparate sources.*
- *Information: the bulk of the course content, my ideas and links to other sources.*
- *Vocabulary: key terms, useful for students to check their own knowledge*
- *Imagination questions: critical thinking and discussion questions*

The sources are linked directly and [most] included in the Bibliography. A glossary and index seemed superfluous, because of CTR-F on most browsers, online dictionaries, and search engines.

I've tried to keep the proportion of material consistent with most introduction to biological anthropology textbooks.

Weeks and Sections

1-2 Introduction

3-4 Evolutionary Biology

5 Cellular Biology

6-7 Primatology

8-10 Paleoanthropology

11-13 Human Variation

14 Conclusion: the Future

One difference in the ordering is that I'm attempting a more chronological approach to human taxonomy than most other textbooks. I introduce human evolution in the order of speciation: from DNA, vertebrates, mammals, primates, hominids, to anatomically modern *Homo sapiens*. I introduce paleontology as the method to understanding primate evolution, and in turn, living primates.

Much of this textbook is just a bunch of links to other articles; almost like a reader. If I found a source that said it better than I could, I just copied the link. I have chosen only articles that are freely available, with no login required. I've tried to include a mix of both the most basic and well-written introduction to the subject, and examples of the most primary and current sources available online. The textbook mixes internet memes, peer review journal articles, encyclopedic entries, video clips, gifs, radio podcasts, games, and VR. There is some justification for a multimodal approach to teaching anthropology (Chin 2017), but there are also drawbacks.

One disadvantage of this reader format is that sometimes students may get bombarded by obscure details. Michelle Field and Tori Saneda describe a similar critique in the introduction to their Wikipedia Biological Anthropology textbook, where they argue for presenting a brief outline of the information online and using class time to fill in the details:

As you peruse the reading material in the course module pages you might find that they contain less detail than what you would see in a "normal" textbook. This is intentional. One thing we find incredible about higher education is that the student often reads the textbook only to go into class and have the professor lecture for two hours on the exact same material. Because of this repetition of the material, students often become exasperated and either stop reading the material or stop paying attention in class. We've also found that students in the introductory anthropology courses frequently struggle with picking out the basic concepts from among the myriad of material from the textbook. We think that students in introductory anthropology courses such as this one, most of whom are not going to be anthropology majors, should read the basic information outside of class. This allows the instructor to focus on providing more explanatory details and help students work through critical thinking about the material in class. Therefore, the readings in the course modules have the basic information. Through in-class activities, discussions, and homework assignments, the job of the instructor is to help students move deeper into and synthesize the material. [Field 2011]

I have tried to address these problems with a short outline to the subjects in my own words, but my textbook emphasizes links to original sources in order to maximize the depth of critical thinking outside of class, and I hope to review the basic concepts during class. Also, repetition is not a waste of time for an introductory class. I believe the risk of losing a few blurry-eyed students to the frustration of obscure websites, is worth the benefits of pushing students to work directly with more primary sources and often the most current research available.

This project is ongoing: links go out of date, better articles become available, permissions are granted, and peer-review is an ongoing process. The quality of a big name textbook is admittedly superior, but I believe that in the context of our community college, overall student success will improve with the mediocre -- but free -- textbook that follows.

Forward

"Can't beat the price!" --Jo Student

Student User Guide

For the [full online version of this textbook](#) I don't recommend printing this entire document because so much of the information is in the hyperlinks that you need to click on to get to. The forthcoming *Study Guide for the Introduction to Physical Anthropology* will be formatted for printing, but in the meantime this LibreTexts version is more printer friendly, then the full version.

It's most efficient to read this with a fast internet connection to minimize the frustration of waiting for links to load. If you don't have a fast connection at home, many browsers have settings that will cache or save pages and let you read them off-line, so you may be able to click all the links here when you have a good connection and then read the pages later.

Experiment with different browsers and try the Reader Mode, you should be able to adjust the font and size so this is not so hard on your eyes.

The WiFi at our college is sketchy, so you should try different locations and times, or find a computer with a direct connection. The new downtown public library has excellent WiFi.

For navigation, the table of contents has hyperlinks to the sections. An index is unnecessary because most internet browsers have a "search" or "find" function within this webpage, trying pressing CONTROL or COMMAND and the letter F at the same time.



Figure 1: "Take Control + F to find your place in the universe" by Arnie Schoenberg, remix of 2014 NASA Hubble photo and 2015 Tracy Watanabe [ctrlF](#) (CC BY-NC 4.0)

This is a work in progress so for any problems you find, please send me an email or let me know in class. If a link is bad you can often find the correct one through a search engine, but please email me the broken link and I will fix it.

I've labeled required links and assignments in GREEN CAPITALS, please consider this part of your required course work; I expect you to do the assignment or click on the link and read it. When I say to skim a link, you want to go to the link with the question of "How is this relevant to physical anthropology?" in mind and scroll thorough quickly looking for answers. If you haven't had much experience skimming before, here is a quick description of skimming scientific articles.

If I just include the link, then you can consider it recommended but not required. Many of the recommend links can be written up as Critical Reviews, which I've labeled with a teal asterisk *. Please consult the instructions for critical reviews in your syllabus for more information.

The Imagination Questions are extra credit assignments that you can answer in a journal format, see your syllabus for details.

Please contact me with specific questions and feedback.

CHAPTER OVERVIEW

1: Introduction to Physical Anthropology

This chapter will introduce you to the field of anthropology, define basic terms and concepts and explain why it is important, and how it can change your perspective of the world around you.

[1.1: Science](#)

[1.2: Anthropology as a Science](#)

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1.1: Science

Science is a specific way of looking at the universe.

science : empiricism :: religion : faith

Anthropology is mostly based on science. Anthropology is holistic. The four main subfields of anthropology are cultural anthropology, physical (biological) anthropology, archaeology, and linguistics. It is worthwhile to back up and introduce both science and anthropology. And before talking about science, we should back up even farther, and talk about *epistemology*, the study of how we know things.

The word science comes from the Latin for “knowledge”, but in modern English it means a very specific kind of knowledge, and implies a method of obtaining knowledge.

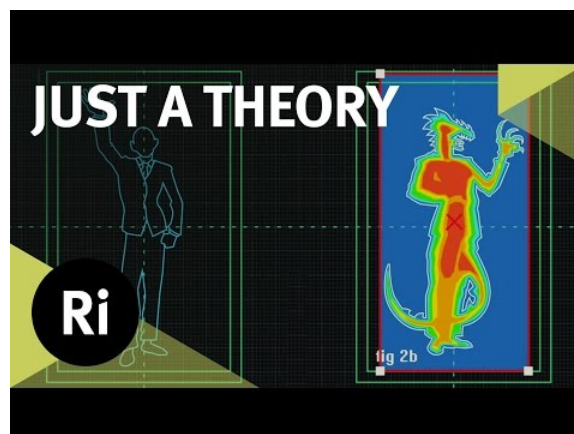
Scientific Method

Note

Another good summary of the scientific method

Here are a few terms to clarify: law vs. theory, quantitative vs. qualitative, inductive vs. deductive

A scientific “Law” is just an archaic term for an accepted theory; we could talk about Newton and his theory of gravity, or Darwin and the Law of Evolution, and what we mean is that neither hypothesis has been disproven yet.



Anthropology uses both quantitative (statistics) and qualitative (detailed description) methods, but leans towards qualitative research.

Other good terms to understand are induction and deduction. Induction is where you take what you can observe and make generalizations, like hypotheses and theories. Deduction is where you start with the general laws of the universe and you use them to predict how a specific event will play out. Both are important aspects of science: the ability to make generalizations, and the ability to predict future events. For example, Sherlock Holmes kept a notebook of all his previous cases, and from this he made inductive generalizations about human nature. When a client came to see him, he would deductively apply his criminal theories to solving the specific case.

Quantitative versus qualitative is another pair to distinguish, and have to do with what kind of data you use. Quantitative science is based on a large quantity of objects, qualitative science is based on intensive scrutiny of a small number of objects. For example, sociology tends to use quantitative methods – studying humans by asking many people a few questions; while anthropology tends to use qualitative methods – studying humans by asking a few people many questions.

All this terminology is relatively new in human history, but the foundation of science, empiricism, is ancient. I don't want to back up too far into philosophy, so let me just say that science is based on what you can experience. Scientists use what you can see for yourself with your own two eyes (or some extension of your eyes, like an electron scanning microscope). If a scientist makes an argument that a fossil belongs to an ancestor of *Homo sapiens*, they need to point out the same details that led them to that conclusion, and as scientists they are required to explain their ideas in a way that anyone else could see the same thing they are seeing and come to same conclusions. This is also an example of how science is "reproducible". Scientists don't get so excited about the **first** person to discover cold fusion, what makes it science is the **second** person to verify the results, or better said, fails to disprove the original hypothesis. Science is about disproving hypotheses. Science can say you're wrong, but it can't say you're right. The goal of science is not about establishing Truth, and for that reason it is often not as satisfying as other branches of knowledge such as religion or art that can claim Truth with a capital "T". Try not to get too frustrated with statements that hedge their conclusions, or just admit that we just don't know yet: this is a characteristic of good science.

Exercise 1.1.1

Read Deduction vs. Induction

How to Critically Review Scientific Articles

Scientific writing can be pretty dry. If you're chatting about a movie, probably the first thing you say is whether you liked it or not, and you'll build up an emotional story without any spoilers.

You need to shift gears for science. This may be disturbing to your self-esteem, but busy scientists usually don't care about how a scientific article makes you feel or whether you liked it or not. They want to know how much the data supports or disproves the hypotheses they are working with. They don't need spoiler alerts, they want as much spoiling as possible. There are more scientific articles than minutes in a human life, so scientists need the conclusion in the first paragraph, so they can decide whether to keep reading or not. The process of doing science entails communicating several necessary components. With good scientific articles you should be able to easily find all of the following elements:

The Citation

A citation is how the article you are reviewing would look in a long list of references, works cited, or in a bibliography. It functions as a link between how you use the ideas in the article, and how the reader can get ahold of the article and read it themselves. It's important to stay anal-retentive about the format of a citation, so people can find the article. MLA and APA are popular formats, but much of anthropology uses the Chicago Style.

The citation functions as the title of your review and goes on top, like the format of an annotated bibliography.

This is the main difference between an essay and a critical review. An essay has a topic and a title, a critical review (or an annotated bibliography) just has a source, and then your thoughts about it below. So what goes below the citation? read on...

The Introduction to Your Review

Even though this is the order of elements in the final version, as you're working on your review, you want to skip ahead and come back here after the Conclusion.

Cover all the following sections and then summarize them into a single paragraph, like the annotation in a typical annotated bibliography, or an abstract that summarizes a longer work. If you organize your review with a paragraph for each section, and each paragraph begins with a topic sentence, then you can pretty much just copy the topic sentences word-for-word and you're done with the Introduction.

Because the introduction is basically a summary of your critical review, you need to do the critical review first, before the introduction.

Why all this jumping around between the article, introduction, and the rest of the critical review?

Your job as the writer is to make it easy for the reader to find the information they need as quickly as possible, and decide if they need to keep reading.

The Hypothesis

Don't waste time! Go straight to the core of the article. What is the author trying to prove? Hypotheses come from “Problems” and “Research Questions” but they are reworded as answers. The hypothesis shouldn't end in a question mark – it's the answer, stated as a concise declarative sentence that is either descriptive (“This is that.”) or causal (“This causes that.”). The question you and the scientists want to answer is how well was the hypothesis supported or disproved.

A Background

Now that you've stated the hypothesis, you can back-up and put it in context. Why is it important? How does the hypothesis connect to other research? For this class you want to refer to the other articles in the same section and explain how this research fits into those broader topics. Why did Arnie put this article in this section? What other sections might it fit into? What does this article have to do with physical anthropology?

This is sometimes called the “Problem” or “Research Question”. In a larger scientific write-up you might make this section into a “Literature Review”, which is a summary of everything that has been written about the subject before your contribution.

The Methods

Now that you've given the essential hypothesis, and given its background, you can add more details about the article itself. What did the scientist do? What techniques or technology were used? How did they look at something? What empirical senses were involved? This is very different from what the scientist thinks—which you find in the hypothesis or the conclusion.

Some Data

If methods are **how** the scientists looked at something, then data are **what** they saw. What did they see? feel? hear? touch? sense? What senses they used are methods, how their senses responded are data. Data are often reported as “Findings” or “Results”.

Scientists are obligated to make their data public so that other scientists can attempt to reproduce their conclusions. An article is already summarizing the data, and a review of an article should summarize it even more. In the humanities and social sciences, hypotheses are rarely completely falsified or overwhelmingly supported, how “true” a hypothesis is, depends on the quality and quantity of the data that supports it.

You want to give enough details to connect the hypothesis to the conclusion.

The Conclusion

The conclusion has two parts: yours and the author's. You want to present the conclusions that the article came to, and you want to wrap your review up, summarizing what you've done so far. In the conclusion, the scientists let us know what they think. How well did the data support the hypothesis? Was the hypothesis testable? Was it reliable (usually a review of the methods)? Was it valid (usually a critique of the background)? Was it verifiable; would it be possible for someone to repeat the same process? How did the sources cited or the “Literature Review” connect to the data? What further research do the scientists suggest?

In a *critical review*, you want to present how the scientists answered these questions. As a *critical review*, you also want to answer these questions yourself, evaluating the strengths and weaknesses of the article. If this were a *peer review* you would be a peer of the author, a colleague, you would have the same background knowledge as the author, and be more likely to thoroughly understand the article and be able to evaluate it. But, you're taking an introductory class, you're not an expert yet, so try not to get cocky and feel like you are supposed to attack the author.

Science thrives on criticism. The goal of a peer review is to help the author fix their mistakes and get better.

Note

Here's a great video of Austin's Butterfly: Building Excellence in Student Work, that reflects the ideal way the scientific community comes to consensus.



Austin's Butterfly: Building Excellence in Student Work from EL Education on Vimeo.

Now that you're done with the body of your review, go back and write a short introduction to YOUR work in the form of an abstract or annotated bibliography.

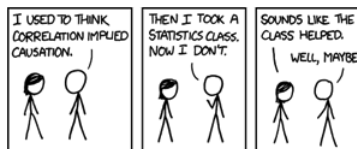


Figure 1.1.2 - <http://xkcd.com/552/> (CC BY-NC 2.5)

- a good source for critically evaluating articles
- Twenty tips for interpreting scientific claims
- Science can't prove anything: how scientists define *theory*, *proof*, *uncertainty*
- The Research Methods Knowledge Base provides a good introduction to these concepts and terms

A Rough Guide to

SPOTTING BAD SCIENCE

Being able to evaluate the evidence behind a scientific claim is important. Being able to recognise bad science reporting, or faults in scientific studies, is equally important. These 12 points will help you separate the science from the pseudoscience.

1. SENSATIONALISED HEADLINES

Aa

Article headlines are commonly designed to entice viewers into clicking on and reading the article. At times, they can over-simplify the findings of scientific research. At worst, they sensationalise and misrepresent them.

7. UNREPRESENTATIVE SAMPLES USED

🔍

In human trials, subjects are selected that are representative of a larger population. If the sample is different from the population as a whole, then the conclusions from the trial may be biased towards a particular outcome.



Figure 1.1.3 - 2014 CompoundChem.com (CC BY-NC-ND 4.0)

Faith

I introduced this section by discussing epistemology, the theory of knowledge. Science is one kind of knowledge, faith is another kind of knowledge. What you know can come from what you experience with your own empirical senses, or you can believe something that someone told you. Faith is complex and varies from person to person, but a concise definition can be found on the popular bumper sticker that reads: "God said it, I believe it, that settles it."

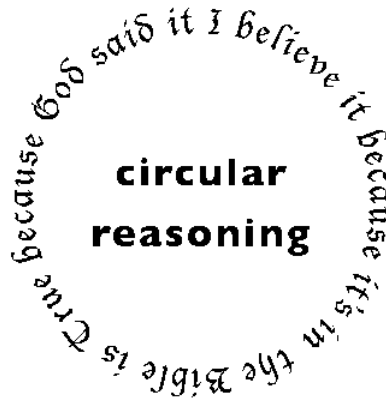


Figure 1.1.3 - Circular Reasoning (CC BY-NC 4.0)

Radical fundamentalist Christianity in the US makes what should be a parlour room discussion between science and religion into a political debate with real educational consequences for all us. Scientists struggle to understand the complex mechanisms of evolutionary theory, but for many, the struggle is made more difficult by ideological barriers set up by faith-based opposition to science.

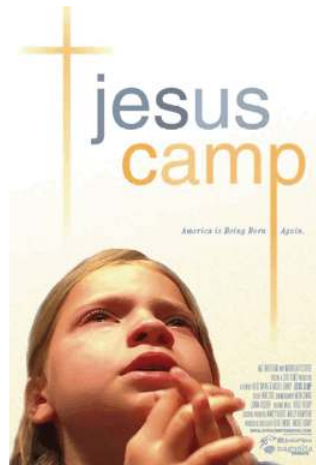


Figure 1.1.4 - Jesus Camp. good 2006 documentary on radical fundamentalism

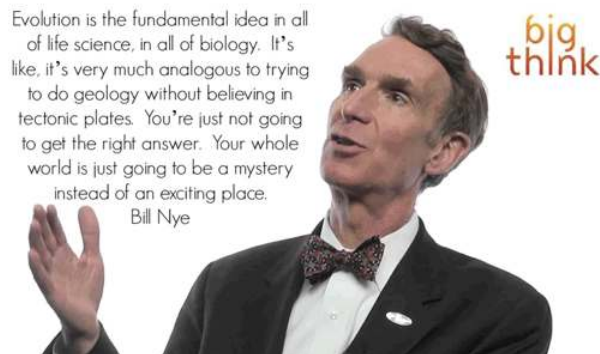


Figure 1.1.5 - Big Think

We need to stress evolutionary theory because it is a fundamental explanatory device in biology. There is a famous quote by geneticist Theodosius Dobzhansky where he states “nothing in biology makes sense, except in the light of evolution.” Trying to make sense of biological systems, including human beings, without evolution is like trying to understand physics while claiming that the force of gravity doesn't exist. There are very few belief systems in the world that deny evolution, and it is totally compatible with most religions around the world, even the Pope has come out supporting evolution (* 2014), so you have no conflict if you are a Catholic. But unfortunately, we happen to live in a culture that prefers faith over science. Our technology has changed rapidly in the last couple of millennia but our mindset has not quite kept up. Many fierce battles are fought today in school boards around the country over the separation of church and state, and whether faith-based ideas (Creationism/Intelligent Design) should be taught in public schools. The battles have spilled out onto the streets with people declaring their beliefs with little symbols on the backs of their cars.



Figure 1.1.6 - Al Seckel, John Edwards 1983

Personally, I wish I didn't have to dwell on the issue so much, I have friends who are Creationists and we get along fine. But, in my role as professor of physical anthropology, I cannot accept Creationism or Intelligent Design as anything more than dangerous fallacies that interfere with students' ability to learn the required curriculum. For me to teach "both sides" would be a form of repressive tolerance (Marcuse 1965). I'm not leading any crusades to banish Medieval thinking from society, but I get worked-up about the issue. It's like professional frustration – that I've failed at my job as an educator – when I see people who are proud of their ignorance. They act like being stupid is somehow cool or something. It reminds me of how Cornel West describes the problem of nihilism in society today.

It bothers me too how religion is used to reinforce socioeconomic class. Many of you are smart enough to transfer to Harvard with a scholarship and get six-figure jobs in the budding genetics industry in major cities around the globe, but if during your job interview, you start spouting Intelligent Design ideology, your scientific credentials go down the drain, and you're back to flipping burgers at In-and-Out.

So, if you think Creationism/Intelligent Design is a load of crap, fine, so do 99.9% of the scientists in the world. But, if you are Christian, you don't have to abandon your faith for this class. I would get in trouble if half the class reasoned, "Well since my teacher has proven that a few poetic lines written thousands of years ago can't be interpreted literally, now I should do the exact opposite of all the moral precepts in the Bible, and become a Satanic mass-murdering tweaker." There are millions and millions of scientists who believe in Christ and evolution and find no contradiction between the two. If that doesn't console you, maybe it'll help to think of this class as an exercise in "know thyne enemy". You don't have to sign a "God is dead!" pledge, and you don't even have to actually believe in evolution to pass this class, you just have to understand it well enough to be able to regurgitate a few of the things I want to hear. But, be forewarned, I'm not a minion of the Devil; it's in my job description to test your faith and proselytize the wisdom of evolutionary theory.

There are very, very few people who actually take The Bible literally. Almost everyone interprets the meaning of the words in The Bible relative to their own language, historical milieu, and personal circumstances. Biblical scholars use the word *hermeneutics* to talk about the different ways that passages in The Bible can be interpreted. For example, there are several passages in the King James' version that says it's harder for a rich man to get to heaven than a camel to go through the eye of a needle. Some Biblical scholars think that the passage was poorly translated from the Aramaic to the Greek, and the mistake continued to Latin, and then English. They have found similar sayings from around that historical period that refer to the difficulty of threading a needle with a camel **hair**, or thread or rope made of camel hair, because camel hairs are so thick. Another interpretation is that Jerusalem had a small gate, called the Camel **Gate**, that was difficult to pass through if you were rich and carrying all your stuff. For me the metaphors of a camel hair through a needle or a rich guy with gear squeezing through a small door make a lot more sense than this image of an actual camel floating through some giant needle. Stuff gets lost in translation. Have you ever played that game, "telephone", where you get a big circle with your friends and one person makes up a complicated sentence and whispers in the ear of the person next to them, who whispers it in the ear of the person next to them, and as the message gets passed around the circle, people miss-hear things, or forget things, and the message changes. And, when you compare the original message to how it ended up, it often sounds silly.

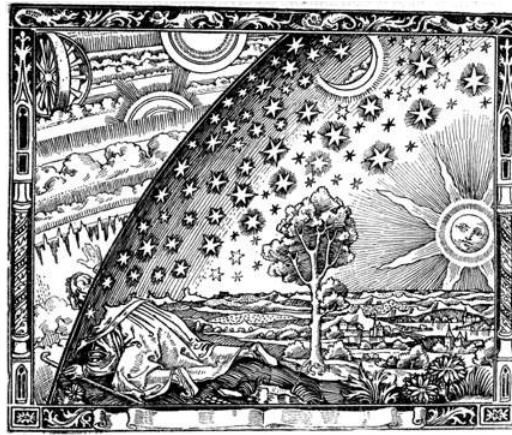


Figure 1.1.7 - Flammarion engraving 1888, notice the pillars that hold up the heavens

Some of the few people who take the Bible literally have been ridiculed into the closet. The Flat Earth Society took literally a passage from The Bible that said the heavens are held up by four pillars at the corners of the earth, and the image of the sun "rising" literally. They actually believed that if you walked far enough to some corner of the earth you would bump into a big pillar. Nowadays, it's hard to find an honest proponent of the Flat Earth theory, but they were around by the 1950s. Now, everyone just laughs at them, like you would probably laugh at someone one who claims the sun revolves around the earth. But on February 17, 1600, this was no laughing matter, when in the Campo di Fiore, Rome, Giordano Bruno was burned at the stake for advocating the heretical belief proposed by Copernicus known as *heliocentrism*.



Figure 1.1.8 - Statue of Giordano Bruno, Campo di Fiore, Rome, Italy (CC BY-NC 4.0)

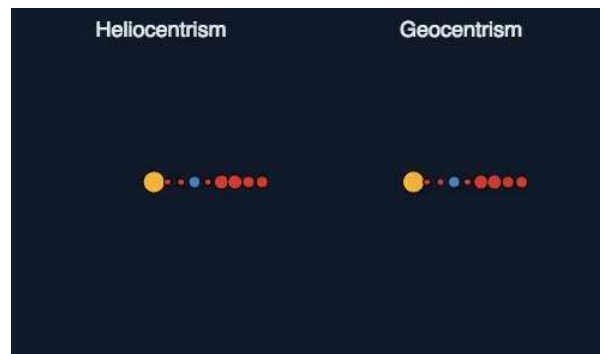


Figure 1.1.9 - <http://i.imgur.com/AReqgfP.gif>

I'm glad that we're at a point in history where I don't have to worry about being burned at the stake for teaching the theory of evolution, and I'm glad that we're at a point where we can just poke fun at Creationists through Darwin fish on our cars, or spoofs such as The Church of the Flying Spaghetti Monster. But I don't think the battle is over. Obama undid many of attacks on science that characterized the Bush administration, but the Trump administration promises a radical attack on science, and an era of "alternate facts". We'll see...

Notes

- A TED talk video of how science can help cure cancer
- A primer on logic and arguments

Vocabulary

- literature review
- abstract
- research question
- epistemology
- anthropology
- archaeology
- biological anthropology
- Creationism
- deduction
- empiricism
- faith
- hermeneutics
- hypothesis
- induction
- Intelligent Design
- laws
- linguistics
- physical anthropology
- qualitative
- quantitative
- science
- taxonomy
- theory

Imagination Questions

- What would happen to science if radical Christian fundamentalists organized a military coup and took control of education in the US?
- What if artificial intelligence advances to the point where robots can do their own science?

- How strong is your belief in science or faith? If you are religious and someone convinced you your religion was wrong, would you give it up? If you believe in science, and you saw God, would you give up your scientific beliefs?

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1.2: Anthropology as a Science

Anthropological Imagination

The anthropological imagination (anthropological perspective) is how anthropologists see the world. Anthropology differs from other sciences because it emphasizes holism and genealogy. The emphasis on genealogy for cultural anthropology implies a focus on the family (domestic structure). The emphasis on genealogy for physical anthropology extends the metaphor of the family tree from an individual and their family, to a family tree writ-large that uses phylogenetic taxonomy to contextualize the human species. Anthropology's emphasis on holism implies a balance between different approaches and many subfields. The four main subfields of anthropology are cultural anthropology, physical (biological) anthropology, archaeology, and linguistics. Anthropologists balance objective and subjective epistemologies.

I think the best way to get a sense of how anthropology differs from other branches of science is to understand the anthropological imagination. I borrowed the concept of the anthropological imagination from one of my professors, Dr. Wade Pendleton, who in turn borrowed it from an introductory anthropology book (Dimen-Schein 1977), who based it on an important sociology book, *The Sociological Imagination* by C. Wright Mills (1959).

There isn't much published about the anthropological imagination, but it is basically the same thing as what most anthropologists call the "anthropological perspective" (Jurmain 2011:19-20; Field 2011). I like the connotations of "imagination" in the way it has been used by John Lennon and recent social movements to recognize the agency that people have to go beyond their cultural constraints. Franz Boas (1858-1942), one of the founders of anthropology, described this as people's need to break the "shackles of tradition" (* *Franz Boas: Shackles of Tradition*). It is especially related to cultural anthropology, where "the world is as you see it", the idea that if people believe in ghosts, then you as a scientist need to start with the hypothesis that those ghosts really exist. That might seem weird to many scientists, but anthropologists need to balance a detached, objective, way of seeing, with the subjective reality of the people they join to study.

Anthropologists balance several seemingly contradictory philosophies. I like to see the anthropological imagination as tendencies between two extreme poles, and though they may lean towards one side or the other, they can never really go to the extreme in any direction.

The Anthropological Imagination Equalizers

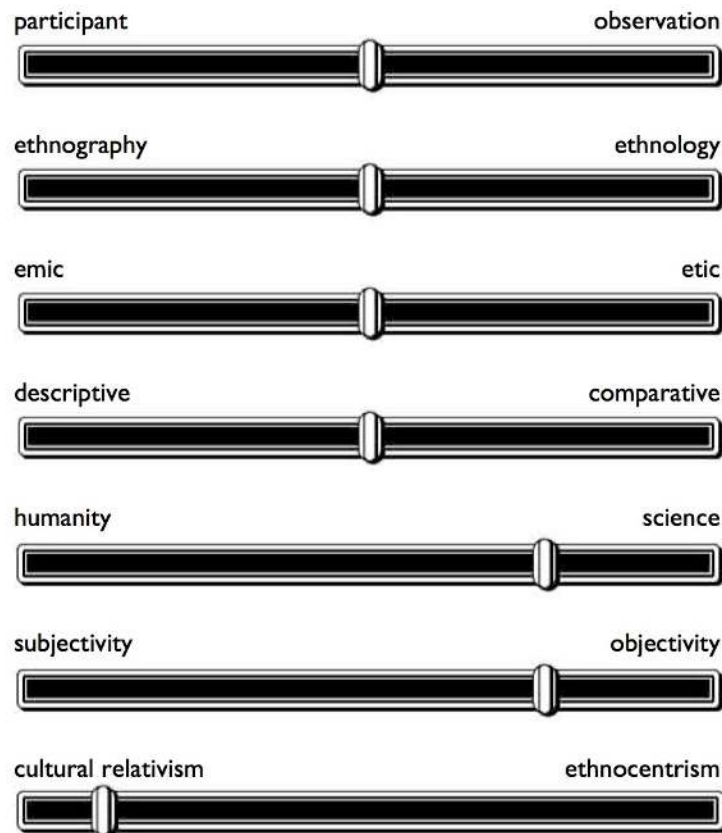


Figure 1.2.1 - *The anthropological imagination balances extremes* (CC BY-NC 4.0)

The principle method of fieldwork in cultural anthropology is called participant-observation, and participant observation exemplifies the balance of anthropology. Anthropologists must objectively study people as an outsider, but they also become part of that culture. They must be culturally relative, and not judge a foreign culture by the standards of the researcher's culture, but they also have their own ethical principles that come from the anthropologist's own culture, and there are limits to how dogmatic anthropologists can be about cultural relativism, and scientists (myself included) need to be a little bit ethnocentric to support things like the *United Nations Declaration of Human Rights and tell another culture that they're doing it wrong. Because our science is so tied to humans we can't avoid asking ethical questions, or as Sir Raymond Firth put it "Cui bonum?" *For Whose Good?* (Schepher-Hughes 1981) What is the purpose of doing anthropology?

This is true to a lesser extent in our class, regarding physical anthropology. We are objectively discussing some aspects of a biological species that has been around hundreds of thousands of years, and has a few distinct characteristics from other animals, but at the same time, we are talking about ourselves, my relatives, the people who gave me the genes I have now, that enable me to think, and type, and wish that this font was easier to read on this crappy screen.

I think the two most distinctive characteristics of anthropology are that it is holistic and it emphasizes genealogy. Holism means that it tries to understand all facets of the human condition. This has many implications. Anthropology is multi-disciplinary, it involves many branches of knowledge. By the early 1900's Franz Boas solidified anthropology into four interrelated subfields: cultural anthropology, physical anthropology, archaeology, and linguistics. As anthropologists began solving real world problems, some advocated for a fifth subfield: applied anthropology. For each of these sub-fields you can combine practically any other branch of science to make sub-sub-fields, depending on your specialization. Don't get too hung up about the correct taxonomy for these branches of knowledge. It can be rewritten in many forms depending which branch you want to emphasize. But in anthropology, all specialists need to have a broad overview to fit their research into the larger questions of what it means to be human, and this incorporation of specific issues into broad questions requires a holistic approach. A good example for physical

anthropology is the concept of *biocultural evolution*, the idea that to understand human evolution we need to look at both biology and culture.

If you take a cultural anthropology class you will see the study of culture requires a holistic approach in its own right because culture is integrated and all-encompassing; you need to study all the elements of culture together and their interaction. Another consequence of holism, and the multi-disciplinary approach of anthropology, is that anthropologists tend to be skeptical of unicausal arguments. A unicausal argument is something like "people have wars because they have an aggressive nature." Anthropologists understand that human nature is supremely complex, and that culture can drastically change any human characteristic that people try to claim is biologically determined. Sure, people are aggressive, you can look at chimpanzees and hominid weapons, but humans are also peaceful, you can look at bonobos and the amazing art of the Upper Paleolithic. Try to keep this point in mind when you're writing for this class: avoid unicausal arguments, give all sides of an issues, avoid oversimplification, explore the evidence that supports each position.

Exercise 1.2.1

Compare ETHNOGRAPY to ETHNOLOGY

Another general emphasis in anthropology is on genealogy. In cultural anthropology, the structure of the family is usually a core element of a culture. In physical anthropology, I like to view our emphasis on classification and taxonomy as just an attempt to better understand the branches of our own family tree.

The anthropological imagination is also something that you as an individual will use to better understand yourself and your place in the world. Biological and cultural explanations can be useful in solving your own problems. Having an answer to "Why am I sweating right now?" means understanding the cooling mechanisms that our ancestors evolved over tens of millions of year, and the fight-or-flight response in response to stress that involves putting your own personal financial problems into a cultural context where education is touted as the method of class mobility, yet restricted by public policy that raises tuition, textbook prices, and limits financial aid.

Subfields of Anthropology

Exercise 1.2.1

Read Dennis O'Neil: The Subfields of Anthropology and Wikipedia's Explanation

Many anthropologists consider applied anthropology as a fifth subfield. I prefer to think of it as a research goal or purpose that cross-cuts all of the four subfields. For example, there are projects that can be considered applied linguistics [language renewal], or applied archaeology [Incan agronomy]. There are many applications for physical anthropology, especially in medicine. The word *forensic* means "legal", but most forensic anthropology tends to be a subfield of physical anthropology - we use what we know about human biology to help solve crimes.

Notes

- The Argentine Forensic Anthropology Team is a good example of forensic anthropology. They have been recently working to identify the 43 missing students from Ayotzinapa, Mexico.
- Check out careers in genetic counseling
- The Health and Medicine chapter from a cultural anthropology textbook
- Desmond Morris' The Language of the Body: a good video combining physical anthropology, cultural anthropology and linguistics

Conclusion: Anthropology and Science

In conclusion, anthropology is mostly a science, but has many aspects of humanism. For a more traditional introduction to anthropology and science read the Dennis O'Neil overview of anthropology.

Epistemology for Physical Anthropology

Epistemology means the study of how we know what we do. *Taxonomy* comes from the Greek word for "branches". Here is a taxonomy of knowledge for this class:

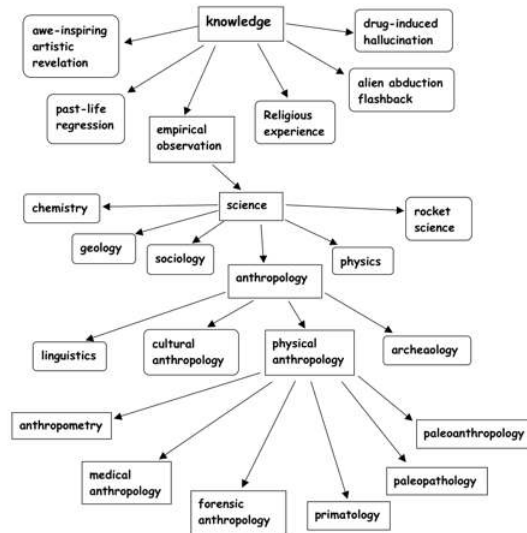


Figure 1.2.2 - An epistemology for physical anthropology (CC BY-NC 4.0)

You add or subtract boxes, and draw the arrows differently depending on what you want to focus on, and because anthropology is holistic these charts don't make much difference; anthropologists include all relevant aspects of knowledge. We will focus on the bottom part of the flowchart, but always keep the bigger picture in the back of your mind.

Vocabulary

- anthropology
- archaeology
- biology
- culture
- epistemology
- ethnography
- ethnology
- linguistics
- participant-observation
- phylogenetic
- taxonomy

Imagination Question

How does the anthropological approach to understanding human beings differ from other classes you've taken?

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CHAPTER OVERVIEW

2: Introduction to Biology

Physical anthropology is also called biological anthropology. You need to understand the basics of biology before you apply it to humans. We are going to use these ideas as the basis for explaining phenomena in later sections, such as: Why primates scream at each other? How are those two fossils related? Why do people look different from each other? What are people going to look like in a million years? Now is a great time to review your notes from your High School biology class.

- [2.1: Introduction](#)
- [2.2: Scale, Human Space, Powers of Ten for Physical Anthropology](#)
- [2.3: Evolutionary Theory \(Part 1\)](#)
- [2.4: Evolutionary Theory \(Part 2\)](#)
- [2.5: Evolutionary Theory \(Part 3\)](#)
- [2.6: "Forces" of Evolution](#)
- [2.7: Genetics, Cellular Biology, and Variation \(Part 1\)](#)
- [2.8: Genetics, Cellular Biology, and Variation \(Part 2\)](#)
- [2.9: Summary Example- Holism in Anthropology, Sickle Cell Anemia and Malaria](#)

Thumbnail: Darwin's finches (also known as the Galápagos finches) are a group of about fifteen species of passerine birds.[1][2][3][4] They are well known for their remarkable diversity in beak form and function. From "Voyage of the Beagle." (Public Domain).

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2.1: Introduction

Physical anthropology is also called biological anthropology. You need to understand the basics of biology before you apply it to humans. We are going to use these ideas as the basis for explaining phenomena in later sections, such as: Why primates scream at each other? How are those two fossils related? Why do people look different from each other? What are people going to look like in a million years? Now is a great time to review your notes from your High School biology class.

Biology is a broad field that ranges in scale from the microscopic to the geographical. Evolutionary theory is crucial to biology. Charles Darwin's theory of natural selection came from a historical context where many other scientists were contributing to evolutionary theory. Natural selection means that individuals compete for resources and those with the variations that make them more fit to survive in a certain environment tend to survive and reproduce more. Sexual selection is the part of natural selection that focuses on competition for reproduction. Darwin understood the importance of variation.

Gregor Mendel used math to explain how variation is inherited. Population genetics used math to show how inheritance works in large populations. The modern synthesis combined Mendelian genetics and population genetics, to codify evolutionary theory into four forces: mutation, natural selection, migration, and genetic drift. Inheritance, mutation, and other sources of variation can now be understood through cellular biology and genetics. Humans are made up of cells. Cells have DNA. DNA is the code that directs protein synthesis. Proteins direct the functions of life.

Sickle cell anemia is an example of how the holistic approach of anthropology can use many subfields, mostly from biology, to understand the origins and consequences of an important human disease.

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2.2: Scale, Human Space, Powers of Ten for Physical Anthropology

Anthropology is a broad field that incorporates many sub-fields and borrows from many other disciplines, so the space that is relevant to physical anthropology is also vast. It ranges from the atomic particle that causes genetic mutation—the primary cause of evolution—to the plasticity of the human body because of our need to adapt to seasonal changes caused by the elliptical orbit of the planet in the solar system. Anything bigger or smaller is beyond the scope of physical anthropology. Why atomic particles behave the way they do is a question of physics. Whether life exists on other planets is a question of astrobiology. Both are good themes for philosophy and science fiction, but are beyond the scope of this class.

Meters	Common Measurement	Abbreviation	Relevance
10^{-10}	1 ångström	1 Å	Size of an atom
10^{-9}	1 nanometer	1 nm	Diameter of DNA helix, size of a base
10^{-8}	10 nanometers	10 nm	Size of a codon (3 base pairs)
10^{-7}	100 nanometers	100 nm	Locus, length of a typical gene
10^{-6}	1 micron	1 μm	Cell nucleus, <i>Plasmodium falciparum</i>
10^{-5}	10 microns	10 μm	Typical primate cell
10^{-4}	100 microns (micrometer)	100 μm	Patch of melanin, width of hair, human egg
10^{-3}	1 millimeter	1 mm	The height of a cusp on a Y-5 molar
10^{-2}	1 centimeter	1 cm	Human ear bones, <i>Anopheles</i> mosquito
10^{-1}	10 centimeters	10 cm	Length of a human's opposable thumb
10^0	1 meter	1 m	Space that an individual primate occupies
10^1	10 meters	10 m	Typical sleeping area of a primate social group
10^2	hectare/ 2.5 acres, football field	100 m	Typical core area of a primate social group
10^3	10 hectares, 1 kilometer	1 km	Typical territory of a non-human primate social group
10^4	100 hectares	10 km	Typical home range of a non-human primate social group
10^5	100 kilometers	100 km	
10^6	1,000 kilometers, continent	1000 km	Typical range of a non-human primate species; range for most of human evolution
10^7	10,000 kilometers, the surface of the planet Earth	100000 km	The range of all biological evolution
10^{12}	1 billion kilometers, 1 terameter	1 Tm	The Earth's orbit in the solar system, includes meteors and solar radiation

Notice that most of the larger ranges are relevant for hominid evolution and primate behavior, the medium sizes are used in osteology and to compare human and hominid variation. The smaller sizes can describe human variation and genetics. Refer back to this chart as we cover those relevant sections.

Exercise 2.2.1

Use this chart and watch the movie POWERS OF TEN



“Recognize that the very molecules that make up your body, the atoms that construct the molecules, are traceable to the crucibles that were once the centers of high mass stars that exploded their chemically rich guts into the galaxy, enriching pristine gas clouds with the chemistry of life. So that we are all connected to each other biologically, to the earth chemically and to the rest of the universe atomically. That's kinda cool! That makes me smile and I actually feel quite large at the end of that. It's not that we are better than the universe, we are part of the universe. We are in the universe and the universe is in us.”
-Neil deGrasse Tyson

Note

More on scale and measurements: human scale

Imagination Question

When you look in the mirror do you see yourself as a trillion cells, or a trillion, trillion atoms which came from stardust, or a collection of selfish genes, or an individual?

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